Image Signal Processing
Structured Syllabus

Unit 1: Image Formation and Geometry

- **Camera Model**
  - Pinhole camera model
  - Camera calibration matrix (K)
  - Intrinsic parameters
  - Extrinsic parameters
- **Geometric Transformations**
  - Euclidean transformation (rotation + translation)
  - Similarity transformation (adds scaling)
  - Affine transformation (general linear + translation)
  - Projective transformation (Homography)
- Perspective projection
- Bilinear interpolation
- **Zoom**
  - Digital zoom (interpolation-based)
  - Optical zoom (lens-based)
- RANSAC (robust homography estimation)

Unit 2: Blur and Focus

- **Blur Types**
  - Space-invariant blur
  - Space-variant blur
  - Gaussian blur
  - Motion blur
  - Defocus blur (disk PSF)

- **Focus Measures**
  - Modified Laplacian
  - Sum Modified Laplacian (SML)

Unit 3: Filtering

- **Noise Model**
  - Additive White Gaussian Noise (AWGN)
- **Spatial (Local) Filtering**
  - Mean filter
  - Gaussian filter
  - Median filter
- **Non-Local Filtering**
  - Yaroslavsky filter
  - Collaborative filtering
  - Block Matching 3D (BM3D)
- Space-variant filtering
- Transform domain filtering
- Image sequence filtering

Unit 4: Image Restoration

- **Forward Model**
$$g = Hf + \eta$$
- **Well-posedness (Hadamard Conditions)**
  - Existence
  - Uniqueness
  - Stability
- **Mathematical Tools**
  - Pseudoinverse
  - Rayleigh quotient
- **Regularization**
  - Tikhonov-Miller regularization
  - Laplacian regularization
  - Stochastic regularization

Unit 5: Estimation Theory

- Mean Square Estimation (MSE)
- Non-linear MSE
- **Optimal Estimation**
  - Conditional mean (MMSE)
  - Law of Iterated Expectation
- **Linear Estimation**
  - Linear MMSE (LMMSE)
  - Normality theorem
- Wiener filter (spatial and Fourier domain)

Unit 6: Frequency Domain and Transforms

- Discrete Fourier Transform (DFT): 1D, 2D, 3D
- Mapping with DFT
- **Data-Independent Transforms**
  - Discrete Cosine Transform (DCT)
  - Walsh-Hadamard Transform (WHT)
  - Discrete Affine Transform (DAT)
- Unitary transforms (energy preserving)

Unit 7: Data-Dependent Transforms

- Principal Component Analysis (PCA)
- Karhunen-Loève Transform (KLT)
- Singular Value Decomposition (SVD)
- Pseudoinverse

Unit 8: Image Enhancement

- Contrast enhancement
- Histogram Equalization (HE)
- Probability intensity transformation

Unit 9: Segmentation and Classification

- Otsu thresholding
- K-means clustering
- ISODATA clustering
- Fisher Linear Discriminant